

Ziegan Wesley

AI ML Engineer / Software Developer

9698099535 zieganwes@gmail.com linkedin.com/in/zieganwesley/ Chennai

SUMMARY

Software Programmer & AI/ML Engineer with a specialized focus on architecting high-performance financial infrastructure and **Edge Vision AI/Gen AI** solutions. Expert in the **ISO 8583 protocol**, building resilient fintech systems and shell-driven automation using **Python, C, PHP, and Bash**. Proven track record in deploying real-time computer vision pipelines on constrained hardware and scaling intelligent solutions via **AWS** and **Azure** cloud ecosystems.

Core Expertise

- **FinTech & Backend Infrastructure:** Specialist in secure transaction switching and backend architecture. Proficient in optimizing **PostgreSQL** databases and streamlining complex financial workflows through custom **DevOps automation** and persistent messaging.
- **Edge AI & Computer Vision:** Extensive experience in the end-to-end ML lifecycle—from data labeling with **Intel CVAT** to training with **NVIDIA DIGITS**. Expert at deploying optimized inference engines using **TensorFlow, PyTorch, OpenVINO, and OpenCV**.
- **Hardware & Real-Time Performance:** High proficiency in leveraging the **NVIDIA DeepStream Platform** to deliver real-time performance on **NVIDIA Jetson (Nano/Xavier/TX1)**, **Google Coral (TPU/Dev Board)**, and **Raspberry Pi**.
- **Cloud & Hybrid Architecture:** Proficient in designing and managing scalable AI and data workloads on **AWS** and **Microsoft Azure**, ensuring seamless integration between edge devices and cloud-based analytical engines.

SKILLS

ISO 8583, Python, C, PHP, Perl, Core Java, TensorFlow, PyTorch, Langchain, ChromaDB, Keras, OpenCV, Scikit-learn, Convolutional Neural Network, Transfer Learning, Model Quantization, CVAT, Nvidia DeepStream, Nvidia Digits, NVIDIA Jetson (Nano/Xavier/Orin), Google Coral TPU, Edge Vision AI, FastApi, PostgreSQL, RabbitMQ, Bash Scripting, Git, Docker, REST APIs, IoT Architecture

EDUCATION

Bachelor of Science, Computer Science

01/2015 - 01/2018

THE AMERICAN COLLEGE

- **Full-Stack Application Development:** Designed and implemented an end-to-end **Hospital Management System** using **VB.NET** within the **Visual Studio** environment, providing a unified interface for healthcare administration.
- **Database Architecture & Management:** Developed a robust backend using **Microsoft SQL Server**, focusing on relational database design, data integrity, and complex query optimization for patient records and billing.
- **Module-Based System Design:** Engineered comprehensive modules for patient registration, doctor availability tracking, and medical history management.
- **Automated Billing & Reporting:** Integrated automated billing logic and pharmacy inventory tracking to streamline financial operations.
- **User Access Control:** Implemented secure login and role-based access levels (Admin, Doctor, Receptionist) to ensure patient data privacy and system security.

KEY ACHIEVEMENTS

Financial Systems Engineering

Scaled and optimized **MRL Posnet (Worldline)** transaction switches by integrating **ISO 8583** protocols for complex payment types and re-engineering **RabbitMQ** messaging to ensure high-availability, real-time settlement for POS networks.

Operational Automation & Infrastructure

Eliminated manual bottlenecks by developing a **Postgres-integrated Bash monitoring suite** that automated transaction flow tracking and EOD settlement verification, drastically reducing operational overhead and system downtime.

Edge AI & Computer Vision Deployment

Engineered and deployed high-stakes deep learning models—including **maritime surveillance and weapon detection**—by performing hardware-specific optimizations for **NVIDIA Jetson** and **Google Coral TPU** to achieve real-time inference on edge devices.

End-to-End IoT & Device Orchestration

Designed and deployed full-stack IoT solutions, ranging from **Micro:bit-based wearable safety prototypes** to **real-time feature activation systems for AI Solutions**, ensuring seamless hardware-software integration and automated smart alerts for diverse environments.

EXPERIENCE

Senior AI ML Engineer

Chennai, India

Cognizant

03/2026 - 05/2026

- Working in Cognizant as a External Client from **Buzzworks Business Services Pvt. Ltd.**
- Using Python, Langchain, ChromaDB for Agentic/Gen AI related WorkFlows and Exploring.
- Creating API Based applications using FastAPI.

EXPERIENCE

Software Programmer

India

Worldline

03/2022 - 12/2025

- Integrated complex payment features by enabling diverse transaction types (SplitPayments, Freecharge, Village Panchayath) on the transaction switch side using ISO 8583 message formats for seamless interoperability.
- Developed real-time notification systems to engineer the communication bridge between the switch and POS devices supporting Soundbox audio notifications for instant transaction confirmation.
- Optimized settlement modules to enhance core settlement logic, ensuring high-accuracy processing of POS transactions within strict, pre-defined time windows.
- Enhanced messaging infrastructure to improve data transmission reliability via RabbitMQ by implementing persistent connections and eliminating forking, leading to a more stable transactional flow.
- Automated operational monitoring by building Bash scripts with Postgres DB integrations to monitor real-time transaction health, identify flow drops, and track EOD settlement status.
- Reduced operational overhead by successfully automating repetitive manual tasks, significantly increasing team efficiency and reducing the risk of human error in critical financial processes.

Junior Software Engineer

Chennai, India

Acceligent Business Solutions Pvt. Ltd

08/2021 - 02/2022

- Real-Time Feature Management: Orchestrated the deployment and activation of POS machine features, enabling seamless functionality updates in real-time.
- System Monitoring: Monitored end-to-end transactional flows to proactively identify bottlenecks or failures within the transaction ecosystem.
- Incident Resolution: Resolved critical transactional flow issues to maintain 24/7 system availability and ensure smooth operational performance for MRL Posnet (Worldline India).
- Cross-Functional Collaboration: Provided technical support as a Junior Engineer at Acceligent Business Solutions, directly contributing to the stability of client-facing payment infrastructure.

AI Software Developer

Chennai, India

ETOP TECHNOLOGIES

12/2019 - 07/2021

- Edge AI deployment: Designed and deployed AI-powered video inference solutions on edge devices, optimizing deep learning models to achieve high-speed, real-time performance.
- Hardware-specific optimization: Focused on model quantization and optimization for specialized hardware, including Google Coral TPU and the NVIDIA Jetson family (Nano, Xavier, Orin).
- Computer vision specialization: Engineered and fine-tuned object detection models for high-stakes use cases, including:
 - Defense & aerospace: Aircraft detection and identification in aerial video feeds.
 - Public safety: Real-time weapon detection for security and surveillance applications.
 - Logistics: Delivery package tracking and recognition systems.
 - Maritime surveillance: Detection of vessels and cargo within port and open-sea environments.
- Full-lifecycle development: Managed the transition of computer vision models from training environments to production-ready edge inference engines.

AI Developer

Chennai, India

YouCode Intelligence Solutions

09/2018 - 12/2019

- Developed a specialized object classification system in Python, optimized for the Google Coral Dev Kit.
- Achieved real-time inference performance by leveraging the Coral TPU for hardware-accelerated deep learning.
- Engineered a safety-focused wearable using Micro:bit, featuring integrated sensors for real-time monitoring and automated smart alerts.
- Architected an IoT-enabled automation system designed for seamless home device control and voice-assisted interactions.